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Moulded-Pulp Packaging: A Straightforward Method for Quickly Designing, Manufacturing and Testing Complex Shapes for Crash Protection Pads

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Abstract: Moulded-pulp packaging stands poised to replace traditional polystyrene packaging due to its eco-friendly nature and recyclability. However, optimizing the protective pins within moulded-pulp packaging has been a challenge, hindering its widespread adoption. This paper presents an innovative approach to swiftly and cost-effectively optimize pin shapes, which is crucial for enhancing impact resistance. We hereby propose a simple try and error pin optimization methodology by utilizing an hand-moulded pulp recipe and PLA 3D-printed moulds to rapidly craft tailored cardboard pins. The validation of the material fabrication was carried out with static characterization and the classification of the homemade moulded-pulp pins with impact testing analysis. This exploratory project demonstrates the feasibility of this approach, laying the foundation for future advancements in moulded-pulp packaging design.

Keywords: moulded-pulp packaging; pin optimization; impact resistance; 3D printing; simulation analysis; eco-friendly packaging

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1. Introduction

Moulded-pulp packaging is emerging as a compelling alternative to conventional polystyrene packaging, offering a sustainable solution that meets the rising demand for eco-friendly products. Composed of natural fibres and fully recyclable, moulded-pulp packaging is not only environmentally responsible but also aligns seamlessly with the goals of a circular economy and greener society [1–7]. One of the main high volume applications of MP packaging is agri-food, with the protection of food (namely fruits) during transportation [8–13]. This type of packaging is fundamentally a thermoformed cardboard sheet that precisely conforms to the shape of the products it encases, ensuring a secure fit that minimizes movement during transportation. In addition to the main form, protective elements such as spikes or pins are integrated into the design, providing cushioning and safeguarding the contents from potential impact damage [14].

However, despite its advantages, moulded-pulp technology faces significant challenges, particularly in the digital optimization of packaging shapes prior to manufacturing [15–19]. This issue is especially pronounced in the design of the protective pins, where the potential for enhancing robustness through more complex geometries remains largely untapped. Those difficulties mainly come from non-linear behaviours of the material (such as rupture) and inhomogeneities within the material, due to the presence of natural fibres. The high costs associated with mould production render traditional trial-and-error methods impractical, forcing designers to rely on empirical approaches that may not fully optimize the design.

To overcome these limitations, we propose an innovative approach that aims to enable the rapid and cost-effective optimization of protective pin geometries. The central concept revolves around the use of 3D printing technology to create individual moulds for a variety of pin designs. These polymer-based moulds are then employed to manually produce

cardboard pins, which are subsequently subjected to rigorous impact resistance testing using a specialized crash test machine. HP Inc. (Palo Alto, CA, USA) has already developed a prototyping tool based on 3D-printed moulds for MP fabrication [20], and we hereby aim at investigating a "quick and dirty" version of such a method in order to make it available to other designers.

This approach not only accelerates the prototyping process and reduces costs but also allows for the empirical testing of innovative pin shapes that might otherwise be difficult to assess. Moreover, the experimental results, such as deformation after impact, can be directly compared with simulations generated by advanced finite element modelling (FEM) tools like ANSYS 2022 R1. By correlating experimental data with simulation outcomes, we can progressively develop a high-performance numerical tool capable of accurately predicting and optimizing the impact resistance of pins with various geometries. In this process, state-of-the-art artificial intelligence techniques may be incorporated to further refine the predictive capabilities of the tool, ultimately leading to more robust and efficient packaging solutions.

The feasibility of this novel approach was tested over the course of a six-month pilot project. This paper presents a comprehensive account of the study's findings, offering detailed insights into the methodology and experimental results: the main outcome of this work is the development of a method aiming at the quick prototyping of new MP shapes for optimized packages, based on standard cardboard material, 3D-printed moulds and crash tests.

2. Materials and Methods

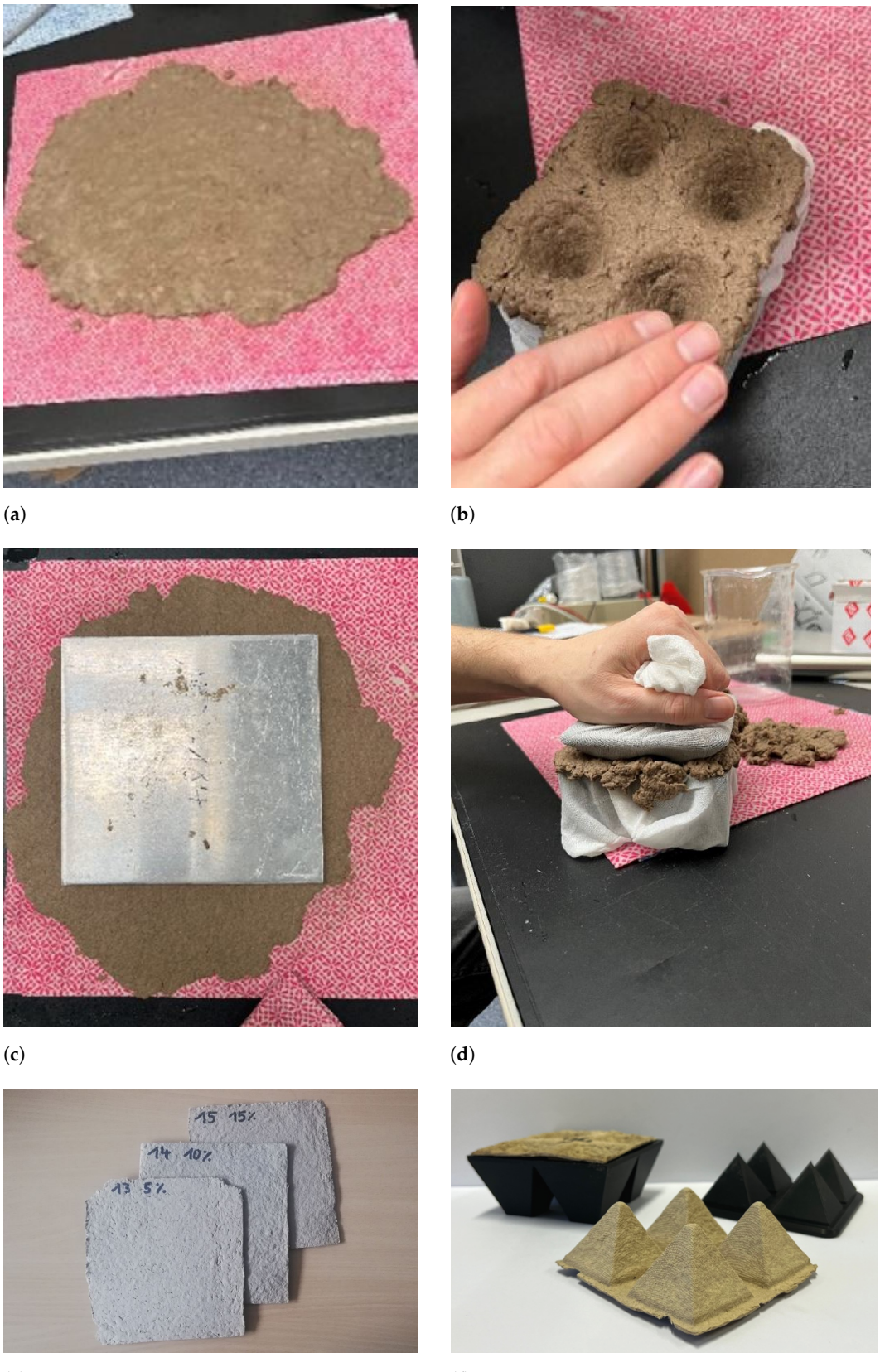
2.1. Fabrication Method

The primary aim of this study is to develop a cost-effective method for rapidly creating and testing new moulded-pulp (MP) geometries without the need for expensive moulds. To achieve this, we devised an economical fabrication process that replicates industrial MP material within a two-day timeframe. Even though a detailed description of this method is provided in Appendix A, a short description of this process is given below:

- Mix cardboard fibres with water;
- Partially dry the paste;
- Insert a specified percentage of white glue;
- Manually place the mixture into a mould;
- Dry it in an oven.

Using this tailored fabrication method, we created various geometries, specifically plates and pins. Static tests were performed on the plates in order to determine the optimal amount of glue needed to ensure that the Young's modulus of our material matched that of the industrial MP cardboard selected for this study. These static tests were performed using a custom-built, 3-point compression test bench. Once the ideal material formulation was established, two distinct pin geometries were designed, fabricated, and tested. The dynamic tests conducted on these samples were carried out using a crash test apparatus.

The MP plates (Figure 1a,c,e) were produced using a roller pin and a square steel plate. Excess material was trimmed, and an absorbent cloth was placed underneath to remove surplus water. The cardboard plate and the steel plate were then dried in an oven for 3 h and 30 min at 80 °C with 5% humidity, identified as the most suitable drying conditions for the preliminary study presented in this paper. Those conditions were selected after many trial-and-error steps. The authors emphasize that they are not aiming at studying or optimizing this drying process, but only to find a suitable one for this preliminary study. The thickness of the plates achieved is around 1 mm.



(**a**) (**b**) (**c**) (**d**) (**e**) (**f**)

Figure 1. Illustration of the fabrication process of the moulded-pulp geometries: plates (**left**) and pyramids (**right**).

The MP pin moulds were designed using Autodesk Inventor, resulting in two geometries: two by two pyramids and two by two cones (Figure 2). The MP pins (Figure 1b,d,f)

were fabricated through a similar process. The pulp was manually placed into the 3D-printed PLA mould (3D printer: Raise3D pro2, Raise3D, Lake Forest, CA, USA, 50% infill) and pressed with the mould's negative part. Both halves of the mould were covered with a net to facilitate demoulding. Finally, the bottom part of the mould and the MP pin were dried in an oven for 6 h at 80 °C with 5% humidity, followed by 15 h at 40 °C with 5% humidity, which were determined to be optimal drying conditions after numerous iterations. The height of these specimens is 30 mm and their width is 100 mm.

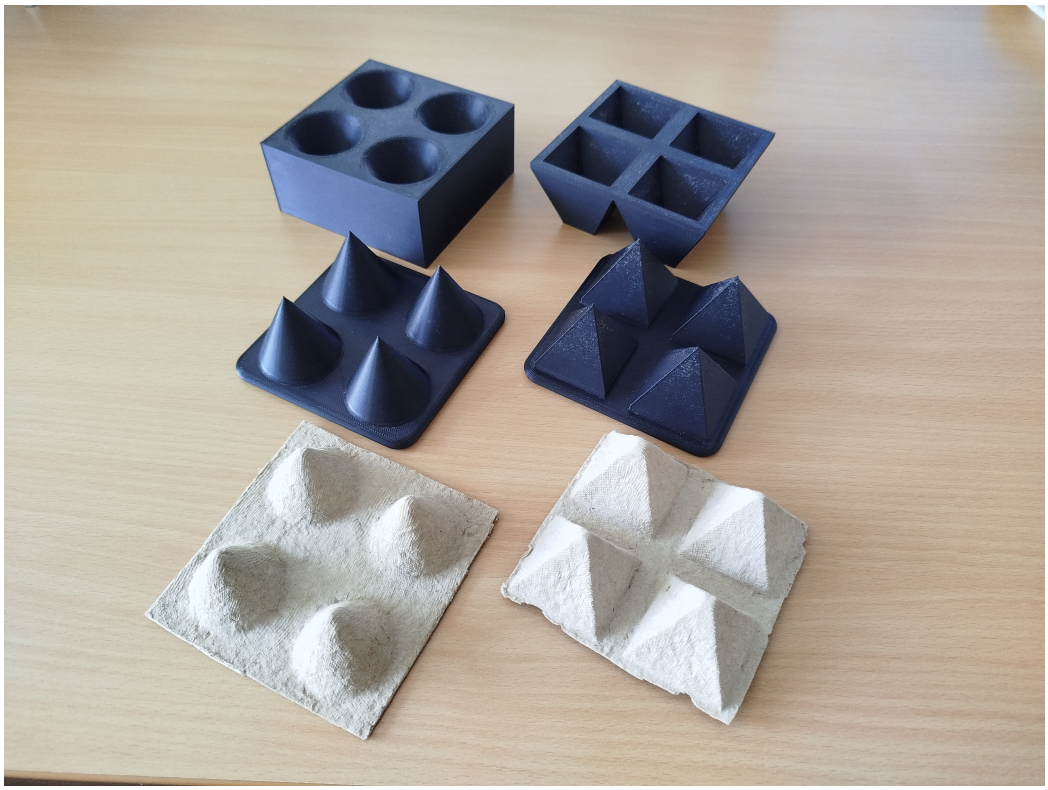


Figure 2. Cones and pyramids: 3D-printed moulds and samples.

2.2. Details on the Static Study Experiments

Three-point compression tests (Figure 3) were conducted on different samples, either cut out from an industrial standard MP package (Figure 4), or from our handmade MP plates.

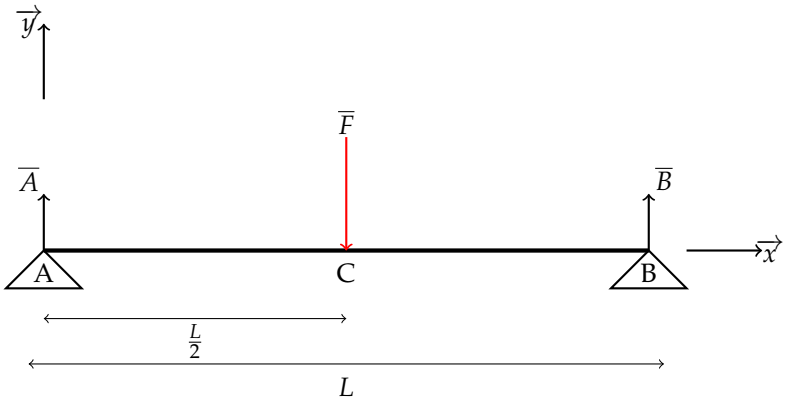


Figure 3. Schematics: 3-point compression of a beam.



Figure 4. Commercial packaging, from which we extracted flat samples for the static study.

With a beam under a 3-point compression load, the maximum deflection at point C, Y_c, is:

$$Y_c = \frac{FL^3}{48EI} \tag{1}$$

where:

- EI is the Flexural rigidity of the beam, with E being Young's modulus and I is the moment of inertia of the beam's cross-section about the axis perpendicular to the applied load.
- L is the length of the beam.
- F is the force applied on point C.

Thus, we obtain the value of the Young modulus from the deformation:

$$E = \frac{FL^3}{48IY_c} \tag{2}$$

The moment of inertia I for a rectangle with width b and height h is given by

$$I = \frac{bh^3}{12} \tag{3}$$

Substituting Equation (3) into Equation (2), we finally get the expression for Young's modulus:

$$E = \frac{FL^3}{4bh^3Y_c} \tag{4}$$

The static test bench, depicted in Figure 5, is constructed from aluminium profiles and PLA 3D-printed components. As shown in Figure 6, the sample is positioned horizontally on a base made of two parallel profiles and placed on a scale capable of measuring forces up to 100 N, allowing for manual recording of the force applied to the sample. A second assembly, also constructed from profiles, is securely clamped to the table and includes a screw mechanism that can be manually turned using a 3D-printed crank. This crank has a hexagonal design, enabling precise adjustments, with each sixth of a full rotation corresponding to a displacement of 0.16 mm based on the pitch of the selected screw. To enhance surface contact with the sample and minimize damage during testing, a 3D-printed half-spherical screw tip is attached. The samples used in the static tests are 95 mm × 30 mm plates, either cut from industrial MP packaging or fabricated using the previously described production method.

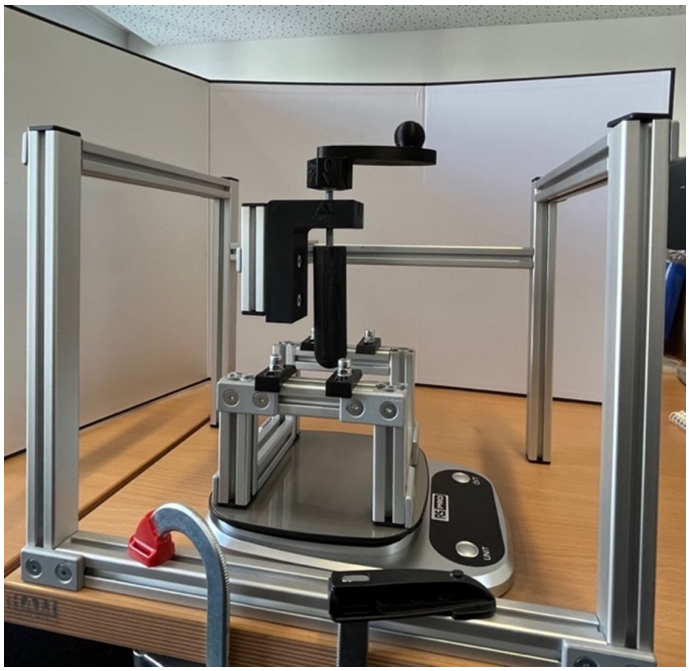


Figure 5. Tailored 3-point compression test bench.

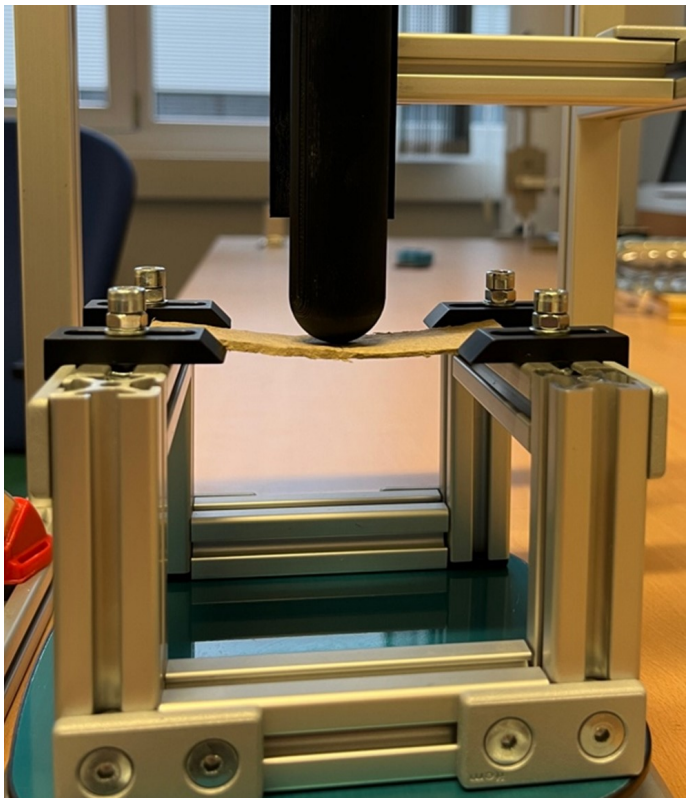


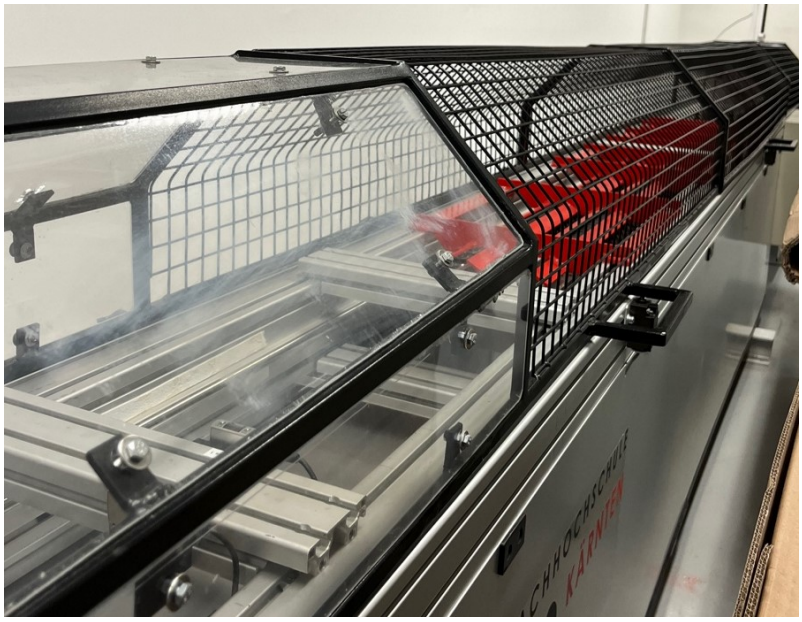
Figure 6. Moulded-pulp plate sample in the compression test bench.

The homemade plates feature different mass percentages of white glue, ranging from none up to 25%, which are used to obtain thin sheets with different Young moduli. Each of those plates is then cut into five samples, and each of these samples is tested using the 3-point compression test bench previously described: by applying a given displacement at the centre of the sample and measuring the resulting force, the Young's modulus can be calculated using Equation (4). These values are then averaged over the five samples extracted from each plate.

2.3. Details on the Dynamic Study Experiments

The objective of this study is to compare the shock absorption capabilities of different MP geometries and have them validated via a simulation. To achieve this, we studied two distinct geometries: cones and pyramids.

The dynamic test apparatus, shown in Figure 7, consists of a launcher that travels horizontally along a track. The launcher is propelled by a spring system, which is first compressed by a motorized mechanism before being released. Upon release, the launcher strikes a grounded vertical plate where the sample is securely mounted using tape (Figure 8). The launcher has a mass of 5.571 kg, and its speed before impact is measured using two optical sensors. This setup allows for the calculation of the kinetic energy transferred to the sample. The launcher was propelled with varying energies by adjusting the compression intensity of the springs, enabling the plotting of a graph displaying kinetic energy versus crushing. The degree of crushing of the sample is determined by measuring the height difference before and after the crash test.



(**a**)



(**b**)

Figure 7. Crash test bench: overview of the track (**a**), launcher and springs (**b**).

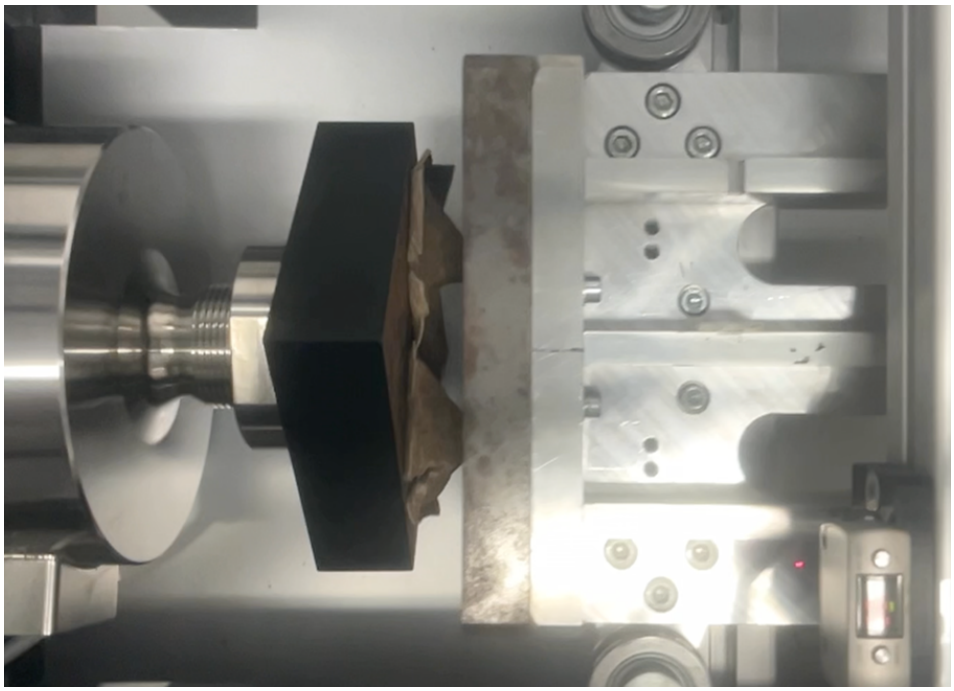


Figure 8. Example of a crash test.

Multiple crash tests were conducted for each geometry (cones and pyramids) at different launcher speeds. However, each of these experiments was conducted a single time, because of the time-consuming fabrication of the MP structures. Examples of crush samples of both types of geometries are shown in Figure 9.



(**a**)



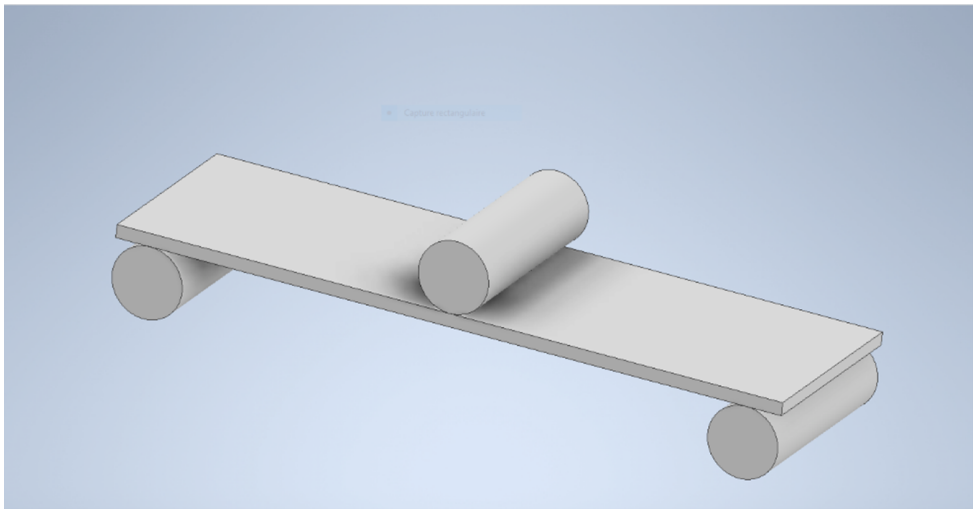
(**b**)

Figure 9. MP pyramid sample after crash (**a**); MP cone sample after crash (**b**).

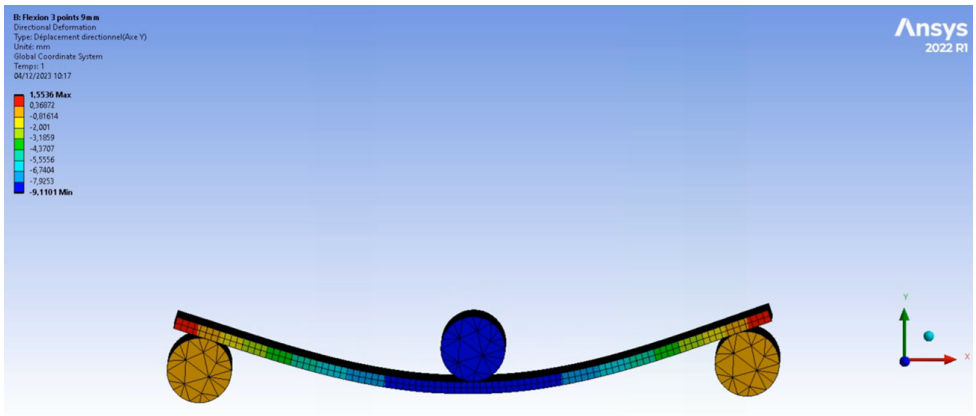
3. Results

3.1. Static Study

The Young's moduli extracted from the different static experiments described in the previous section are input into a numerical model created in ANSYS (Figure 10) using structural mechanics to verify the consistency of these experimental results. The cylinders are rigid bodies and their contacts with the sample is frictionless. The two external cylinders are fixed while the centre cylinder is moving down the vertical axis. The meshing elements are tetrahedral.



(**a**)



(**b**)

Figure 10. Simulation example of a MP sample under a 3-point compression test: geometry (**a**) and calculated deformation along the vertical axis, in mm (**b**).

Due to the fabrication process, the thickness of each plate varies slightly. Therefore, in order to visualize a linear relationship between the applied displacement Y_c and the resulting force F, the force normalized by the sample thickness raised to the third power $\frac{F}{h^3}$ versus Y_c is plotted, instead of F vs. Y_c (see Equation (4)). Figure 11 shows the comparison between experimental and simulation results for each dataset.

Two conclusions can be drawn out of the graph in Figure 11. Firstly, both experimental and simulation results are consistent, which confirms the methodology used here to extract Young's moduli of the different plates. Secondly, one of the handmade plates exhibits similar mechanical properties than the industrial MP plate. It concerns the one containing 25% of white glue. We therefore selected this formulation as the most suitable one to fabricate our MP geometries, including the ones made for the following dynamic studies.

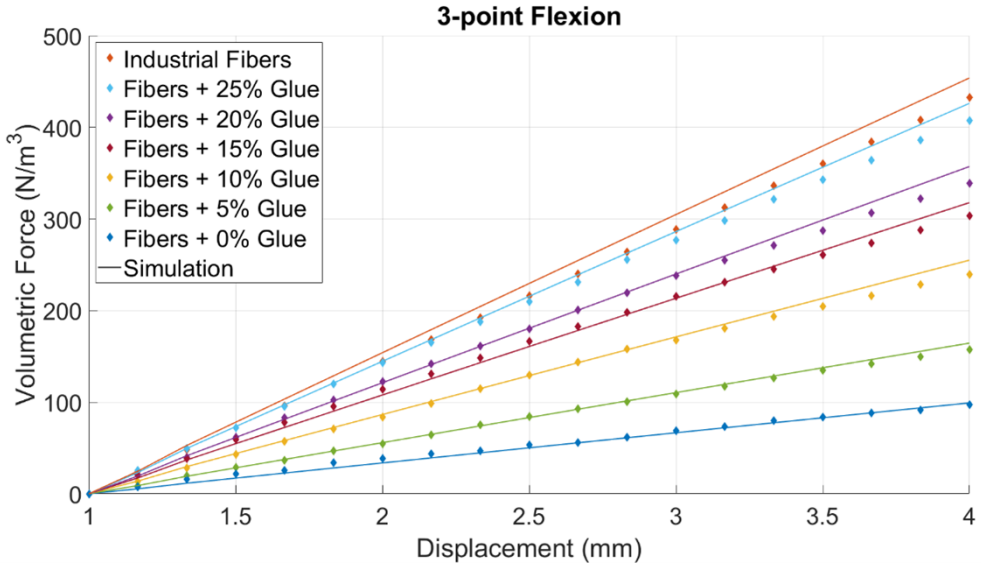
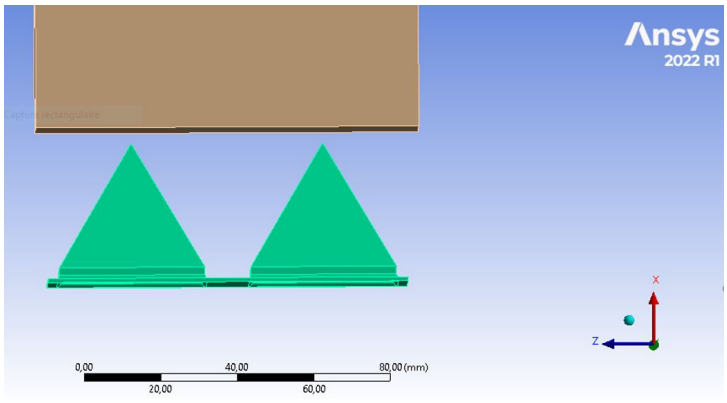


Figure 11. Results of the static study for various percentages of white glue: simulation vs. experiments.

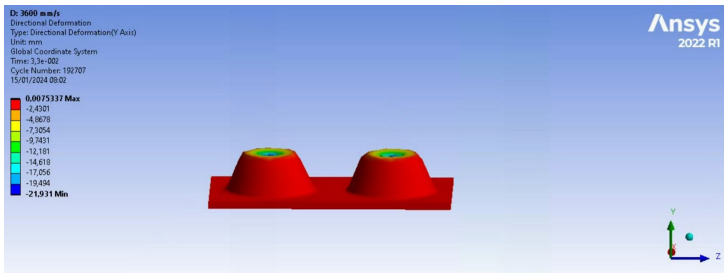
3.2. Dynamic Study

A numerical model was developed using the explicit dynamics module in ANSYS. The body of the MP structure (in green in Figure 12a) is deformable, while its bottom surface is fixed. The launcher is made of steel (in brown in Figure 12a), and the meshing elements are tetrahedral. Young's modulus of the MP material was set to 692 MPa, based on our results from static experiments. However, Poisson's ratio was assigned a standard value of 0.3 since no experiments were conducted to determine this parameter. Other factors, such as non-linearities and material rupture, were also not considered, and this must be taken into account when interpreting the results. Nonetheless, the primary purpose of these simulations was not to precisely replicate the experimental outcomes but only to check if the ranking of the geometries according on their shock absorption efficiency is the same than that based on the experimental data (i.e., which geometry is more effective in absorbing a given shock). Examples of simulated crush for both types of geometries are shown in Figure 12b,c.

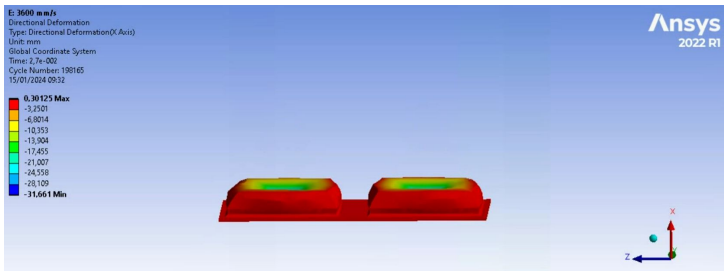


(**a**)

Figure 12. *Cont.*



(**b**)



(**c**)

Figure 12. Crash tests simulations: model (**a**); cones (**b**) vs. pyramids (**c**); same launcher velocity (3.6 m/s, which corresponds to an impact energy of 36 J). The deformations displayed are on the vertical axis.

The results from both simulations and experiments are plotted in Figure 13. The results indicate that the pyramidal geometry crushes more easily than the conical geometry under a given impact energy. This could be attributed to the fact that pyramids have sharper angles, where stress is more concentrated, leading to plastic deformation and subsequent mechanical failure in those regions. As a result, the pyramidal geometry exhibits a lower capacity to absorb high impact energies. For instance, at 60 J, the pyramid is already fully crushed, corresponding to the sample's height of 30 mm. Conversely, in the conical geometry, the impact forces are more evenly distributed, resulting in a higher absorption capacity before failure. The simulations exhibit a similar trend, with the pyramids crushing more than the cones under the same impact energy. However, the difference observed between the two geometries is more pronounced in the simulations, likely because the back plate on which the samples are mounted was not modelled. This omission leads to simulated crushing values that exceed reality, even surpassing 30 mm, which is impossible in the experiments.

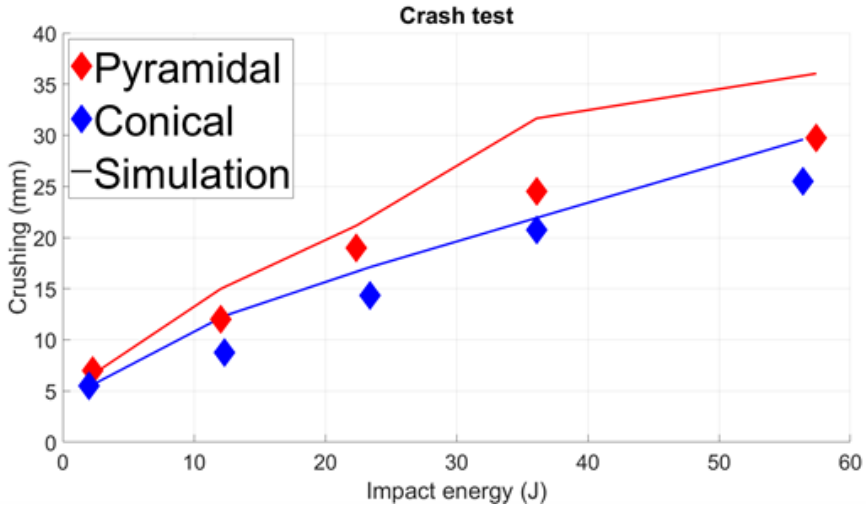


Figure 13. Dynamic study results for various launcher speeds: experiments vs. simulations.

4. Discussion

In this work, we developed a method that can be implemented in almost any R&D lab for fabricating custom moulded-pulp geometries. The material used, a blend of cardboard and white glue, was engineered to exhibit mechanical properties similar to those of industrial MP structures. This method allows for the classification of geometries based on their ability to absorb shocks. The setup is straightforward, requiring only a standard 3D printer, an oven, and a crash test bench. It enables the evaluation of multiple geometries within a two-day period. First, we manually fabricated and dried moulded-pulp plates from the cardboard/glue mixture. Static tests were conducted on these samples to identify the optimal proportion of white glue required to match the Young's modulus of industrial MP structures. In the next phase, we used PLA 3D printing to create moulds of various shapes, and devised an efficient method for correctly positioning the material within the moulds and drying it in an oven. These steps led to successful crash tests, the results of which were validated throughout FEM simulations.

Namely, cones and pyramids pins were tested. Such geometries were also tested years ago in [21], leading to the conclusion that both structures have very close mechanical dynamic behaviour. However, this study used expanded polystyrene foams instead of paper. In our study, we could show that conical shapes have a better dynamic strength than pyramidal ones, results that are consistent with the ones from [22], where similar geometries were crash tested. Finally, the study carried out in [7] demonstrates that structural factors such as the sample geometry have a higher influence on the shock absorption capabilities than the material properties such as density or thickness. This result confirms the importance of optimizing the shape of the MP packages, which is what our method tends towards.

The approach presented in this study effectively demonstrates the feasibility of fabricating and roughly characterizing tailored moulded-pulp pin geometries with relative ease. By using this method, significant advancements can be made in accelerating the prototyping capabilities for optimizing MP pins in industrial applications. However, despite the promising results, several challenges and limitations remain that need to be addressed for this approach to become more broadly applicable and efficient.

- Fabrication challenges with complex geometries: Certain geometries are inherently difficult to fabricate using this method, particularly those involving vertical walls. The vertical edges present a significant challenge because they complicate the unmoulding process, leading to potential damage or deformation of the structure upon removal from the mould. For example, a cylindrical geometry was manually fabricated as part of this study, but the process was not entirely successful. The manual approach introduced inconsistencies and defects, highlighting the limitations of the current fabrication method when applied to more complex or vertically oriented designs.
- Time-consuming fabrication process: The entire fabrication process currently spans two days: the first day is dedicated to pin design and 3D printing, while the second day is required for material preparation and the subsequent drying phase. This timeline is not optimal for industrial applications where rapid prototyping is essential. The drying phase, in particular, is time-consuming due to the necessity of slowly heating the pulp to remove excess water.
- Surface quality and structural integrity: The presence of excess water in the pulp mixture during the moulding process poses another limitation. Because the water content prevents the mould from being fully closed during drying, the resultant structure tends to have a smooth surface only on one side. This asymmetry can lead to inhomogeneities within the material, creating localized weaknesses that could compromise the structural integrity of the final product. More generally, the effect of any remaining humidity in paper packages on their structural integrity is a topic that is currently of major interest [23], namely because of the storage conditions of such packages. Defining a fabrication process with better controlled humidity is therefore a major future possible improvement for this work.

To overcome these challenges, one potential solution involves the design and utilization of advanced moulds with channels that allow water to escape. By incorporating these channels, it would be possible to use a vacuum chamber to draw out a portion of the water from the pulp mixture before the final drying stage. This process would allow the structure to harden more evenly, enabling the structure to be fully unmoulded before drying and resulting in a smoother and more uniform surface on all sides of the geometry. Moreover, by pre-hardening the structure in this manner, the overall time required for oven drying could be reduced, leading to a shorter fabrication cycle and higher-quality outputs. However, this approach would necessitate more resources and increased design complexity due to the need for more sophisticated mould engineering. The design and manufacturing of such moulds would likely require more time, as they would need to accommodate the additional channels and vacuum features.

In addition to improving the fabrication process, the dynamic characterization of the fabricated MP geometries could be conducted using a simplified test bench. The current setup relies on springs to generate impact forces, but a more straightforward vertical drop test could achieve similar results with less complexity. By dropping weights along a vertical track onto the samples, rough characterization of the shock absorption capabilities of different geometries could be performed. This method would be sufficient for a comparative analysis if the primary goal is to rank the effectiveness of the geometries relative to one another. Such a test would reduce the need for more elaborate setups while still providing valuable insights into the performance of fabricated structures.

Finally, our ANSYS model of the dynamic tests remains relatively basic and could be improved to reach higher accuracy, for example, by refining the mesh in critical regions of the geometries such as in the angles and at the tip of the structure, like in more developed MP packaging crash test simulations [24]. Plasticity and other non-linearities such as rupture should also be taken into account to provide more accurate results.

5. Conclusions

This brief feasibility study presents a streamlined method for the rapid prototyping and testing of various moulded-pulp (MP) geometries. This study outlines a straightforward, accessible process for handcrafting MP pin structures using readily available commercial materials, namely white glue and cardboard, in combination with standard 3D printers and ovens. This approach is designed to quickly optimize MP geometries.

This study goes beyond fabrication to explore the practical potential of these MP structures by demonstrating a simple yet effective method for classifying the geometries based on their performance in crash tests. By conducting controlled impact experiments, this study provides a preliminary assessment of the shock absorption capabilities of different MP geometries. This classification serves as a proof of concept, highlighting the method's utility in evaluating and optimizing MP designs for industrial applications. By refining the fabrication process, improving mould design, and simplifying the characterization methods, this technique can be optimized to meet the demands of fast prototyping and provide a reliable basis for the development of tailored MP products with enhanced performance characteristics. Ideally, such experimental results can also serve as a reference to numerical models in order to improve them until they are able to predict the crash test output.

In summary, this study not only offers a practical recipe for creating MP geometries but also establishes a foundational approach for their preliminary evaluation, paving the way for more advanced research and development in this area.

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Appendix A. A Recipe for Handmade Paper Pulp

In this appendix, we describe in details the paper pulp fabrication process used in this study.

The primary ingredient in the manufacturing process is industrial cardboard, which undergoes several steps to transform it into hand-moulded pulp structures. Initially, it is cut in two to three centimetres pieces and then immersed in water for approximately twelve hours in order to dissociate the fibres of the cardboard (Figure A1a). Subsequently, the soaked cardboard is blended (Figure A1b) before undergoing filtration and draining to remove excess water (Figure A1c,d), resulting in a dried fibrous material. This dried material is then weighed (Figure A1e). White glue, crucial for binding the fibres, is then also weighted based on a specific percentage relative to the weight of the previously measured dried fibre pack (Figure A1f). Next, these dried cardboard fibres are reintroduced to water and agitated to form a paste-like consistency (Figure A1g). This paste is then gently drained to keep it soft (Figure A1h). Finally, both the glue and the cardboard paste are thoroughly mixed to ensure homogeneity (Figure A1i).

The mixture is then poured or shaped into moulds, which define the final form of the packaging. Finally, the moulded pulp is subjected to a long drying process, involving temperatures from 70 °C to 90 °C, to dry but not cook the material.



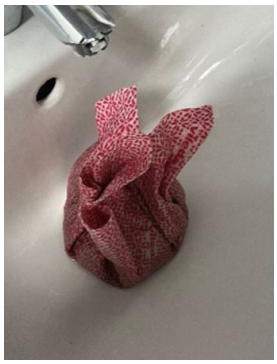
(**a**) Cardboard cut and soaked.



(**b**) Soaked cardboard mixed.



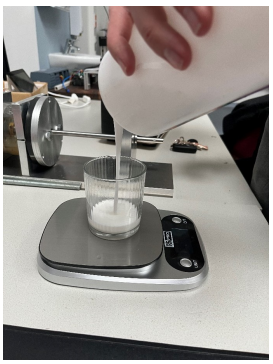
(**c**) Mixed fibres drained.



(**d**) Mixed fibres dried.



(**e**) Dried fibres weighted.



(**f**) Glue weighted.

Figure A1. *Cont.*



(**g**) Dried fibres re-mixed.



(**h**) Fibres gently drained.



(**i**) Wet fibres and glue mixed.

Figure A1. Cardboard pulp fabrication steps.

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